

PROJECT PROPOSAL

Parallel Histogram Equalization using High Performance Computing

EC7207 - High Performance Computing

Submitted By: Group No 18

Fernando K.M.K.N.C. (EG/2021/4508)

Fernando L.N.K. (EG/2021/4510)

Fernando S.A.U. (EG/2021/4511)

1. INTRODUCTION & PROBLEM STATEMENT

Histogram Equalization is a fundamental image processing technique used to enhance contrast by redistributing pixel intensities based on the Cumulative Distribution Function (CDF). This project aims to accelerate Histogram Equalization by implementing and evaluating four distinct architectural approaches: **Serial**, **Shared Memory** (OpenMP), **Distributed Memory** (MPI), and **Hybrid** (MPI + CUDA). The study will analyze the trade-offs between computational speedup and communication overhead.

2. OBJECTIVES

- **Establish a Baseline:** Implement a verified Serial version to serve as the “Gold Standard” for accuracy.
- **Implement Parallelism:** Develop optimized solutions using **OpenMP**, **MPI**, and **Hybrid MPI+CUDA**.
- **Evaluate Performance:** Measure execution time, Speedup ($T_{serial}/T_{parallel}$).
- **Verify Accuracy:** Quantify data integrity using Root Mean Square Error (RMSE).
- **Analyze Scalability:** Assess performance scaling across varying thread counts and image resolutions.

3. METHODOLOGY

- **Phase 1: Serial Implementation (Reference):** A standard single-core CPU implementation will read the image, compute the global histogram, and map new pixel values. This output will verify the correctness of parallel versions.
- **Phase 2: Shared Memory (OpenMP):** The image will be decomposed row-wise. To prevent race conditions during histogram calculation, threads will utilize local private histograms which are later aggregated via a reduction clause.
- **Phase 3: Distributed Memory (MPI):** The image will be partitioned across multiple nodes, each computing a local histogram. These will be combined to form a global cumulative distribution function (CDF), which is then applied to reconstruct the final enhanced image.
- **Phase 4: Hybrid (MPI + CUDA):** This approach targets large-scale datasets, where each processing node offloads its image segment to a GPU. Parallel kernels will compute histograms using optimized shared memory techniques to efficiently handle high-concurrency updates while minimizing memory contention.